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ImagiChem: Deterministic Image-Conditioned Generation of Chemically Valid and Drug-Like Molecules

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ImagiChem: Deterministic Image-Conditioned Generation of Chemically Valid and Drug-Like Molecules

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ABSTRACT: Generative artificial intelligence has significantly expanded the accessible chemical space; however, the systematic use of visual information as a conditioning signal for molecular generation remains largely unexplored. Here, we present ImagiChem, a deterministic, image-conditioned molecular generation framework that maps visual features to chemically valid, drug-like structures via a formal, rule-based assembly process. Local color distributions, gradient statistics, and texture descriptors are deterministically encoded into a constrained set of atomic, functional, and topological transformation rules, defining molecular connectivity without stochastic sampling. The resulting molecules consistently satisfy physicochemical and medicinal chemistry constraints, including Lipinski and Veber criteria, low PAINS liability, and favorable synthetic accessibility. Across extensive image inputs, ImagiChem generates over 99% structurally novel molecules relative to major chemical databases while maintaining controlled

distributions of molecular weight, polarity, and ring complexity. Perturbation analyses of the input images demonstrate that localized visual variations selectively modulate scaffold topology while preserving global property profiles, indicating a high degree of structural controllability and reproducibility. ImagiChem introduces a complementary paradigm to data-driven generative models, enabling interpretable, deterministic exploration of chemical space conditioned on visual information, and provides a reproducible framework for integrating computer vision descriptors into molecular design.

INTRODUCTION

Drug discovery remains one of the most complex, time-consuming, and costly endeavors in biomedical research. The development of a single approved drug can take more than 12 years and cost more than 1 billion USD, with only 1 molecule in 10,000 synthesized reaching the market.^{1,2} Despite remarkable advances in computer-aided drug design (CADD), the discovery process still depends heavily on incremental optimization of known scaffolds and rational exploration of limited chemical regions.^{1–3} Traditional structure-based CADD (SB-CADD) approaches, such as molecular docking and molecular dynamics simulations, have been accelerated by breakthroughs in structural biology, including the explosion of experimentally determined or AI-predicted protein structures from tools like AlphaFold.⁴ Yet, the vastness of the chemical universe, estimated at up to 10^{60} possible small molecules,^{1,5} renders exhaustive search infeasible, necessitating new paradigms for molecular exploration.

De novo drug design (DNDD) has emerged as a transformative strategy to address this challenge. DNDD methods aim to design new compounds from scratch, leveraging heuristic or generative

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3 algorithms to efficiently traverse the immense chemical space.¹⁻³ The recent rise of deep
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5 generative models, including recurrent neural networks (RNNs), variational autoencoders (VAEs),
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7 transformers such as MolFormer and ChemBERTa, and diffusion-based molecular generators, has
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9 provided powerful means to simulate molecular creativity.⁶ However, even the most sophisticated
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11 architectures remain inherently biased toward known data distributions, as they learn from human-
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13 curated chemical datasets (e.g., ChEMBL, ZINC, PubChem). Consequently, their latent spaces are
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15 constrained by precedent, and the probability of generating truly novel chemotypes that are remote
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17 from the training data is small.⁵
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21 This limitation highlights a paradox in modern drug design: as computational models become
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23 more rational and efficient, they may also become less serendipitous. Yet, serendipity has played
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25 a pivotal role in the history of pharmacology. Landmark discoveries such as penicillin, identified
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27 by Alexander Fleming through a chance contamination of a Petri dish,⁷ sildenafil's unexpected
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29 vasodilatory effect (Viagra),⁸ and the analgesic properties of aspirin derived from plant
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31 metabolites⁹ all underscore the power of unplanned discovery. True innovation often arises when
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33 randomness intersects with structured reasoning,^{1,10} a concept that computational pipelines
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35 typically exclude by design.
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39 To reintroduce algorithmic serendipity into molecular generation, we sought an unconventional
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41 form of input, rich in structure yet detached from chemical precedent: visual art. Artworks encode
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43 complex patterns of composition, symmetry, color distribution, and form, reflecting multi-scale
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45 organization akin to molecular diversity. By translating these aesthetic features into chemical space
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47 via deterministic, hierarchical mapping, it becomes possible to traverse regions of molecular space
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49 that conventional models rarely explore.
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Here, we introduce ImagiChem, a generative algorithm that converts digital artworks into chemically valid molecules. ImagiChem is fully deterministic: given the same image, it consistently produces the same molecular set. Its purpose is not to replicate known QSAR trends or medicinal chemistry heuristics, but to act as a translational bridge between aesthetic information and molecular representation. In doing so, it introduces controlled randomness, an “engineered serendipity”, that may lead to unconventional yet valid scaffolds and unexpected chemical architectures.

This paper presents the conceptual and computational foundations of ImagiChem, describes its algorithmic pipeline for converting visual inputs into molecular outputs, and demonstrates its application to a series of historical painting artworks.

RESULTS

The ImagiChem pipeline

ImagiChem’s design comprises four deterministic stages (Figure 1), each of which contributes both reproducibility and interpretability:

- (i) deterministic seed derivation and image preprocessing,
- (ii) pixel statistics to a chemical alphabet mapping,
- (iii) hierarchical molecular assembly,
- (iv) finalization and evaluation.

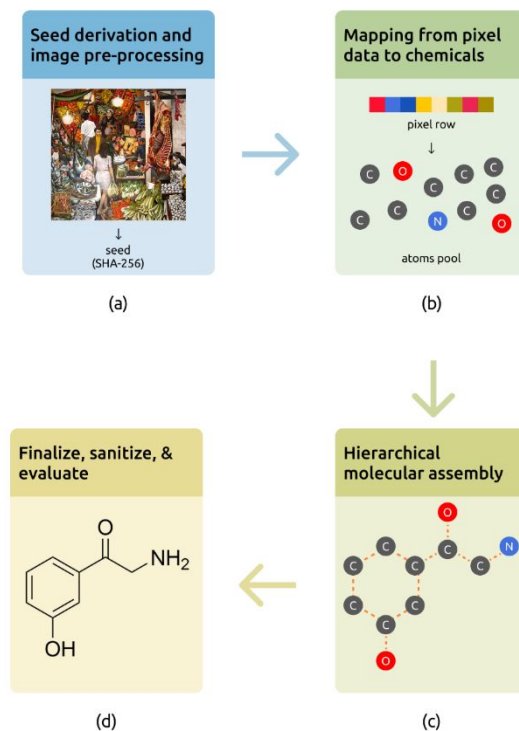


Figure 1. Overview of the ImagiChem pipeline. Stage (a) converts image data into a deterministic seed; (b) encodes pixel statistics into a chemical alphabet; (c) performs hierarchical molecular assembly; and (d) validates and ranks generated molecules by chemical and pharmacological metrics.

Deterministic seed derivation and image preprocessing. At the core of ImagiChem lies the principle of deterministic reproducibility: given the same input artwork, the system must always yield the same molecular output. At the same time, even the slightest visual alteration produces a distinct result. This is achieved through cryptographic seed derivation and controlled image preprocessing.

Each run begins by converting the input image to grayscale, reducing its dimensionality to a single luminance channel. This focuses analysis on contrast and texture, the structural elements

most closely tied to compositional form, while ensuring cross-comparability among artworks with differing color palettes. By operating solely on luminance values, ImagiChem mirrors the perceptual mechanisms that emphasize light–dark relationships as the foundation of visual structure.

From the grayscale image, the first row of pixel values serves as the deterministic anchor for the generative process. These values are passed through the SHA-256 cryptographic hash function,¹¹ which computes a 256-bit digest that serves as the seed for all random number generators (RNGs) used throughout the pipeline. Owing to the avalanche property of SHA-256, even a single-pixel modification results in a radically different hash, guaranteeing that the corresponding molecular generation pathway diverges entirely.

The derived seed initializes all RNGs across Python’s random, NumPy, and custom stochastic libraries, ensuring that every probabilistic decision, from atomic assignment to structural branching, is pseudo-random yet reproducible.¹² Thus, ImagiChem achieves a balance between sensitivity and determinism: a computational analogue of creative variation, where microscopic changes in visual input cascade into distinct chemical narratives while preserving full reproducibility within each unique artistic configuration.

Pixel statistics to chemical alphabet mapping. Each row of pixel intensities in an image is treated as a one-dimensional signal, from which ImagiChem extracts descriptive statistics (mean, variance, contrast, and peak count) that are then heuristically mapped to a chemical alphabet. This process, termed data molecularization, transforms aesthetic or visual signals into structured chemical grammars capable of generating plausible molecular compositions.

The mean intensity of a row determines the number of carbon atoms, reflecting how the central tendency in light distribution corresponds to the formation of an organic backbone. The standard deviation, which expresses variability around the mean, is associated with the abundance of nitrogen atoms, introducing polarity and chemical diversity. The global contrast of the row controls molecular size: high contrast implies greater visual richness or informational complexity, which is reflected in larger, more structurally complex molecules. Meanwhile, the peak frequency, which is the count of local maxima along the row, governs the sulfur content, symbolizing local irregularities and functional diversity.

A pixel is recognized as a peak only if it satisfies three local conditions: (i) its intensity is greater than that of the preceding pixel, (ii) greater than that of the following pixel, and (iii) exceeds a dynamically computed threshold based on the intensity range of that specific row. This adaptive criterion ensures that peaks represent meaningful variations rather than noise.

From these mapped parameters, an extended molecular formula is constructed, a string of atomic symbols such as CCNCO..., whose elemental composition mirrors the statistical behavior of the pixel data. To maintain pharmacological plausibility, halogens (e.g., F, Cl, Br, I) are restricted to $\leq 5\%$ of the total number of atoms, preserving realistic proportions found in drug-like molecules.

In essence, this mapping establishes a consistent and interpretable bridge between *visual structure* and *chemical semantics*, allowing aesthetic features in an image to emerge as quantifiable molecular blueprints.

Hierarchical molecular assembly. Molecular construction within ImagiChem follows a hierarchical, chemistry-inspired assembly process, designed to emulate the structured reasoning of synthetic and medicinal chemists rather than relying on random atomic concatenation. The system

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3 begins by selecting a foundational molecular core from a curated fragment library of over 1,500
4 single-pharmacophore fragments (i.e., discrete small molecules each bearing a single binding-
5 capable functional moiety, such as a polar group) from Enamine.¹³ In parallel, the algorithm
6 chooses from a set of predefined cyclic or polycyclic scaffolds (e.g., benzene, pyrrole, naphthalene
7 rings, etc.) drawn from substructures commonly found in marketed pharmaceuticals.¹⁴ These are
8 selected for their chemical stability, favourable synthetic tractability, and known relevance to
9 bioactive small molecules. Such scaffolds serve as the chemically robust foundation upon which
10 subsequent molecular elaboration is built, ensuring that the initial hit fragment already embodies
11 structural validity, synthetic accessibility, and pharmacological relevance.
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24 The algorithm dynamically selects a core whose atomic requirements can be satisfied by the
25 available atom pool, which is itself generated from the preceding pixel-to-chemical mapping stage.
26 Once a suitable scaffold is established, ImagiChem initiates an iterative expansion phase, in which
27 substituents or individual atoms are attached according to probabilistic rules guided by the
28 composition of the remaining atom pool, valence constraints, and heuristic structure–activity
29 relationships (H-SAR). Each addition step respects chemical bonding rules and explores functional
30 diversity while maintaining overall molecular coherence.
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40 To prevent dead ends in molecular growth, the system continuously monitors the valence of all
41 atoms. Suppose no attachment points remain, but additional atoms are available. In that case, the
42 algorithm may strategically relax bonding constraints, for example, converting a double bond into
43 a single bond, to reopen possible sites for substitution. This flexibility allows ImagiChem to
44 explore a wide variety of chemically valid yet unconventional configurations, striking a balance
45 between structural realism and creative diversity.
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In effect, this structure-aware, compositional hierarchy mirrors the aesthetic logic of artistic layering: stable frameworks (the core scaffolds) provide form and balance, while successive modifications introduce complexity and nuance. The result is a computational analogue of compositional harmony, in which molecular architectures emerge through guided, hierarchical refinement, faithful to both the principles of synthetic chemistry and the creative dynamics of artistic construction.

Finalization and validation. The final stage of the ImagiChem pipeline converts internally generated molecular graphs into standardized, chemically validated representations suitable for further analysis and application. Each molecular graph is transformed into an RDKit¹⁵ object, leveraging the library's robust cheminformatics infrastructure to enforce chemical correctness. Through the Chem.SanitizeMol routine, ImagiChem verifies and corrects valence states, bond orders, and aromaticity patterns, ensuring that each structure conforms to real-world chemical rules before proceeding to evaluation.

After sanitization, each valid molecule undergoes pharmacological evaluation, which assesses Pan-Assay Interference Compounds (PAINS),¹⁶ i.e., compounds that tend to yield false positives during High-Throughput Virtual Screening (HTVS). Furthermore, the molecules' synthetic accessibility (SA) is quantitatively evaluated by assigning a synthetic difficulty score ranging from 1 to 10^{17} . A score of 1 represents molecules that are very easy to synthesize, while a score of 10 indicates molecules that are highly challenging to synthesize. Prioritizing molecules with lower SA scores enables ImagiChem to efficiently filter and classify thousands of generated compounds based on their chemical synthesizability. This automated assessment streamlines the selection

process by focusing on candidates with favorable synthetic feasibility, thereby accelerating downstream experimental efforts.

To further refine the molecular set, duplicate structures are detected and removed via canonical SMILES comparison, ensuring each retained compound is unique in the output space. The resulting collection of molecules, now fully validated and ranked, is exported in a .csv file containing the SMILES, SA, and PAINS columns. From this file, all SMILES codes can be easily extracted into a .smi file for downstream computational screening, visualization, or synthesis planning.

In this final stage, ImagiChem unites computational precision with chemical rigor, transforming aesthetic, signal-derived abstractions into a curated portfolio of chemically meaningful, pharmaceutically plausible molecular entities.

Application to artistic works

To assess the creative and pharmacological potential of ImagiChem, the system was applied to a corpus of ten celebrated artworks (Table 1) spanning diverse periods, styles, and visual structures, enabling ImagiChem to be tested across a wide range of aesthetic conditions. Renaissance works such as *The Birth of Venus*, *Mona Lisa*, and *Madonna of the Meadow* offer balanced compositions and harmonic geometry. At the same time, modern pieces like *Starry Night* and *The Scream* introduce expressive contrasts and high visual entropy. Paintings rich in symbolism and tonal variation, such as *Vucciria* and *Badende Frauen*, further test the algorithm's sensitivity to complex visual inputs. Together, this curated selection ensures that ImagiChem's performance reflects genuine responsiveness to differences in form, color distribution, and luminance, validating its generality across distinct artistic and structural paradigms. All paintings were retrieved from the

Artvee website (<https://artvee.com/>) to ensure high-resolution input images. Only *Vucciria* and *Virgin Annunciate*, which we wanted to include for personal emotional reasons, were downloaded separately (<https://www.theworldofsicily.com/consigli-di-viaggio/12-quadri-famosi-da-vedere-in-sicilia/>) as they are not available on Artvee.

Each painting was processed under identical deterministic conditions, enabling direct comparison of their molecular outputs and the aesthetic–chemical relationships encoded within.

Across all images, ImagiChem generated between 675 and 4,699 molecules per artwork, depending on resolution and luminance complexity (Table 1). Every molecule was subsequently cross-referenced against the PubChem database¹⁸ to determine novelty. On average, 99.17% of the generated molecules were previously unreported, confirming the system’s ability to consistently explore unexplored regions of chemical space regardless of image dimensions or stylistic origin. This extraordinary rate of molecular innovation highlights ImagiChem’s role as an engine of aesthetic-driven chemical creativity, an algorithmic analogue of artistic inspiration rendered in molecular form.

Remarkably, among the minority of rediscovered molecules, one compound, 3,3-diethylpyrrolidin-2-one (DEABL, PubChem CID 9793807), emerged from Antonello da Messina’s *Virgin Annunciate*. DEABL is an approved anticonvulsant drug,¹⁹ and its spontaneous regeneration from a purely visual stimulus demonstrates ImagiChem’s ability to converge, through aesthetic data alone, on pharmacologically meaningful regions of molecular space. This finding underscores the system’s capacity for algorithmic rediscovery, in which art functions not merely as inspiration but as structured input that guides chemical reasoning.

Table 1. Molecular Diversity Generated by ImagiChem Across Ten Artworks

Artwork (Artist)	H	W	Tot mols	New	Existing	NF%
<i>Bacchus</i> (Annibale Carracci)	1500	1316	1500	1483	17	98.87
<i>Bathing Women</i> (Ernst Ludwig Kirchner)	1452	1800	1452	1448	4	99.72
<i>Madonna of the Meadow</i> (Raffaello Sanzio)	4699	3669	4699	4662	37	99.21
<i>Mona Lisa</i> (Leonardo da Vinci)	1800	1226	1800	1770	30	98.33
<i>Starry Night</i> (Vincent van Gogh)	1391	1800	1391	1360	31	97.77
<i>The Birth of Venus</i> (Sandro Botticelli)	1307	2094	1307	1303	4	99.69
<i>The Scream</i> (Edvard Munch)	1800	1351	1800	1796	4	99.78
<i>The Wedding at Cana</i> (Paolo Veronese)	1718	2528	1718	1715	3	99.83
<i>Virgin Annunciate</i> (Antonello da Messina)	675	506	675	664	11	98.37
<i>Vucciria</i> (Renato Guttuso)	675	692	675	675	0	100.00
TOTAL	17017	16982	17017	16876	141	99.17

Drug-likeness and pharmacological evaluation

To assess the pharmacological plausibility of the molecules generated by ImagiChem, a comprehensive drug-likeness analysis was conducted using three independent predictive models: (1) the QED from RDKit, (2) the DBPP indices implemented in DBPP-Predictor,²⁰ and (3) the Druglikeness and Drugscore indices implemented in DataWarrior.²¹ Each model provides complementary insights into how closely a compound resembles known bioactive molecules in terms of physicochemical and structural properties.

Prior benchmarking against a reference dataset of 2,342 FDA-approved drugs (from MedChemExpress) revealed that the DBPP-Predictor offers the most reliable discrimination between drug-like and non-drug-like compounds (Figure 2). Approximately 63% of approved drugs in this validation set exceeded the DBPP threshold value of 0.736, identified by the model's authors as the cutoff for favorable drug-likeness.²⁰ This result justified the adoption of DBPP as the principal metric for ImagiChem's pharmacological evaluation.

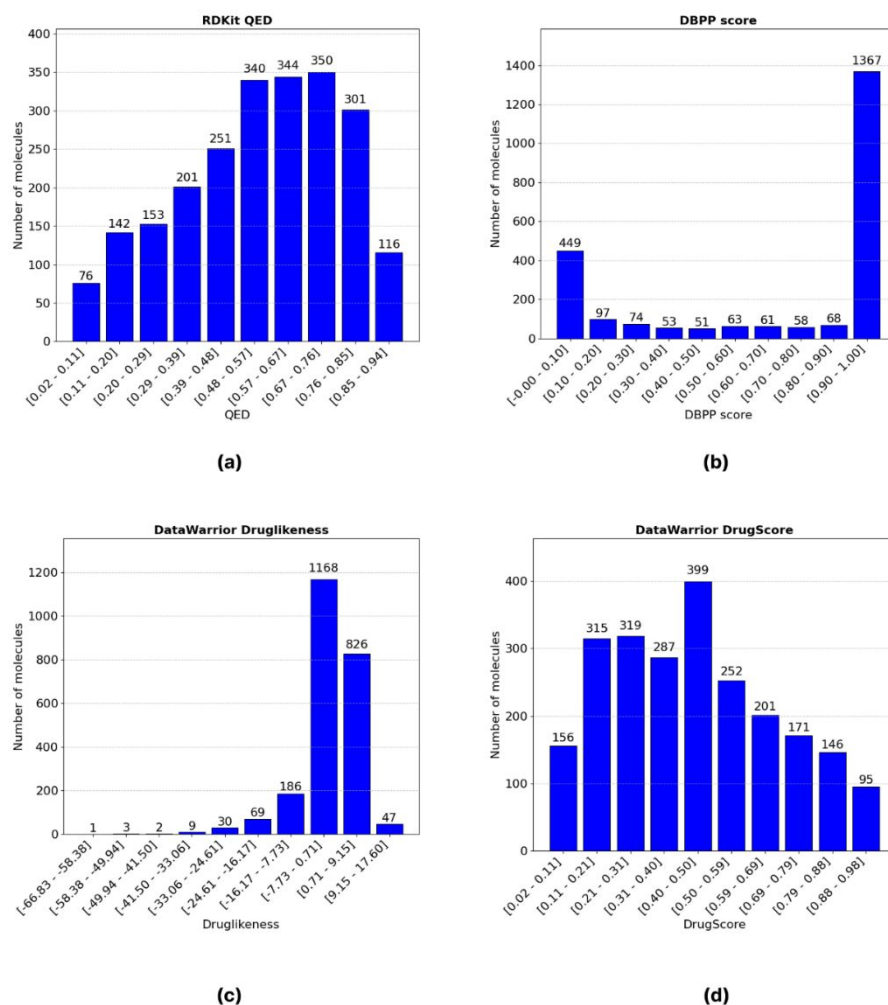


Figure 2. Graphs showing the results of druglikeness analyses performed by the QED RDKit-based implementation of Biscu-it(tm) (a), DBPP scores from DBPP-Predictor (b), Druglikeness (c), and Drudscore (d) from DataWarrior. For each graph, the y-axis shows the number of

molecules, and the x-axis shows the value ranges, where square brackets indicate that the extreme value is included in the range. Molecules given error during RDKit's QED calculation weren't included.

When applied to the complete molecular set generated from the ten analyzed artworks, ImagiChem achieved a mean of 13.55% of molecules exceeding the DBPP threshold (Table 2). This represents a remarkable enrichment given that the algorithm operates without any chemical priors or training bias toward known pharmacophores. The distribution of high-scoring molecules was heterogeneous across artworks: for instance, *The Birth of Venus* (Botticelli) yielded the highest proportion (28.77%), whereas *Starry Night* (van Gogh) yielded the lowest proportion (2.80%). Such variation reflects the influence of each painting's visual entropy and structural composition on the resulting chemical diversity.

Table 2. Drug-Likeness Distribution Across Artworks Based on DBPP Threshold (0.736)

Artwork	Total molecules	DBPP score >0.736	% Above Threshold
<i>Bacchus</i>	1500	192	12.80
<i>Bathing Women</i>	1452	221	15.22
<i>Madonna of the Meadow</i>	4699	704	14.98
<i>Mona Lisa</i>	1800	63	3.50
<i>Starry Night</i>	1391	39	2.80
<i>The Birth of Venus</i>	1307	376	28.77
<i>The Scream</i>	1800	159	8.83
<i>The Wedding at Cana</i>	1718	413	24.04
<i>Virgin Annunciate</i>	675	19	2.82

<i>Vucciria</i>	675	120	17.78
TOTAL	17017	2306	13.55

High-DBPP compounds often originated from images with rich color contrast and textural density, suggesting that aesthetic complexity correlates with increased molecular intricacy and aromaticity. Representative examples from *The Birth of Venus* and *Madonna of the Meadow* are shown in Figure 3, illustrating the translation of visual richness into pharmacologically relevant molecular architectures.

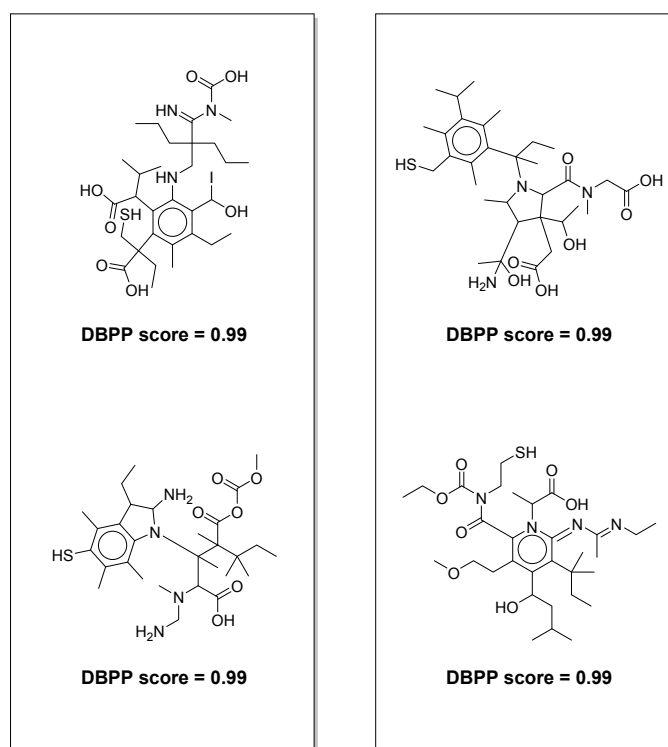


Figure 3. Representative high-DBPP molecules generated by ImagiChem from Botticelli's *Birth of Venus* (left panel) and Sanzio's *Madonna of the Meadow* (right panel). The aesthetic features (color contrast and texture density) correlate with molecular complexity and aromaticity.

Inverse Virtual Screening (IVS) and target prediction

To assess the biological relevance of the aesthetic-driven molecular generation process, the ten molecules (Figure 4) with the highest DBPP scores obtained from Guttuso's *Vucciria* were subjected to IVS. The analysis followed a previously established validated workflow by our group,²² employing the GNINA docking engine to predict ligand–target affinities through convolutional neural network scoring (CNN_VS). Each ligand was screened against a curated panel of 26 experimentally validated protein targets spanning GPCRs, kinases, and enzymes (Table S1), representing a wide range of pharmacological classes.

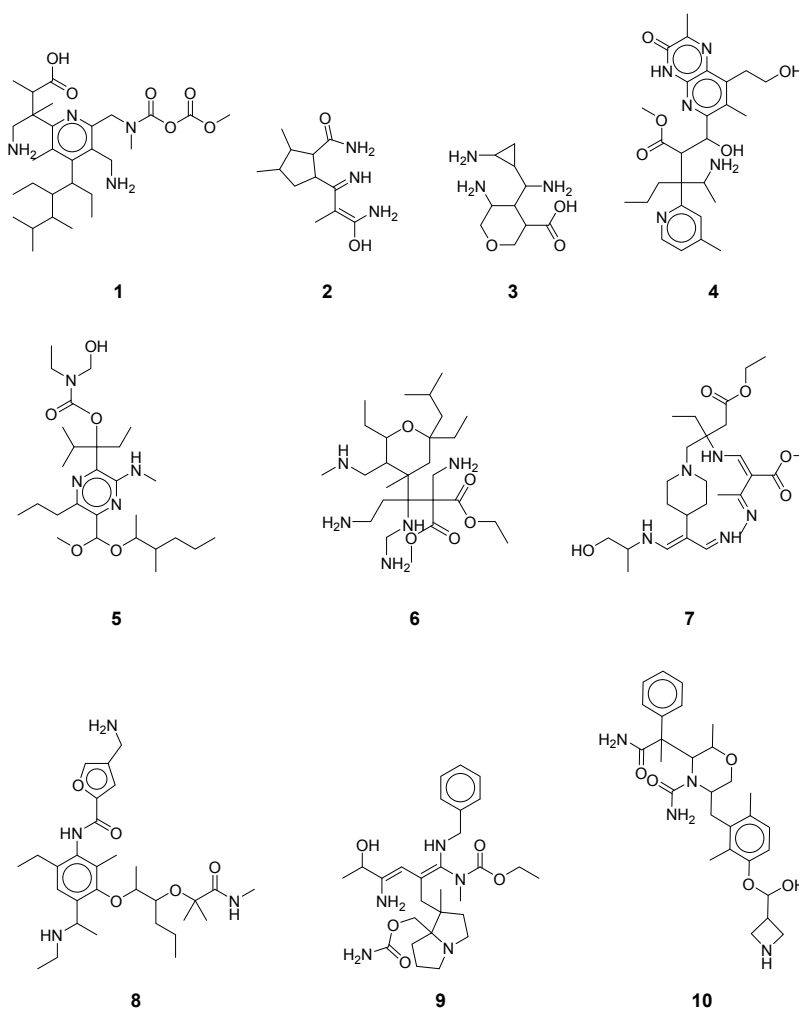


Figure 4. Representative two-dimensional structures of the top 10 ImagiChem-generated compounds tested by IVS.

The results, reported in Table 3, demonstrated that several art-derived molecules exhibited predicted binding affinities above the CNN_VS threshold of 6.3, indicating a substantial likelihood of a meaningful ligand–receptor interaction. Notably, compound **4** showed high affinity for the adenosine A_{2A} receptor, **5** exhibited interactions with HSP90 α and the β_1 -adrenergic receptor, **7** demonstrated dual potential for muscarinic M3 and PPAR γ receptors, **8** revealed multi-target behavior, engaging acetylcholinesterase, dopamine D3, and muscarinic M3 receptors, and **10** displayed predicted activity toward acetylcholinesterase.

Table 3. IVS Results for the Ten Compounds with the Highest DBPP Scores Obtained ImagiChem-Generated Molecules from Guttuso’s *Vucciria*. Each Heatmap Cell Represents Predicted Ligand–Target Affinity (CNN_VS Value). Molecules Exceeding the 6.3 Threshold are Marked in Green, Indicating Potential Pharmacological Relevance

Compound	1	2	3	4	5	6	7	8	9	10
Target name										
Adenosine A _{2A}	2.91	3.64	0.71	7.10	5.69	1.18	2.42	4.69	2.01	4.33
Tyrosine-protein kinase ABL2	1.88	2.61	1.40	3.61	1.79	0.49	1.54	2.05	0.53	4.76
AKT2	2.60	2.37	2.53	2.64	1.39	0.85	2.73	2.73	1.52	3.87
Acetyl-CoA carboxylase 2	1.79	1.56	0.93	2.19	4.31	0.92	1.88	1.32	2.90	2.63
Acetylcholinesterase	3.86	3.67	3.70	5.40	4.72	4.91	5.74	7.83	3.63	6.39
BACE-1	4.85	4.23	2.07	4.51	4.61	3.59	2.04	3.12	5.51	4.72

Carbonic Anhydrase II	1.54	2.31	0.42	2.83	3.42	0.36	1.38	1.10	1.18	1.34
Cyclin-dependent kinase 2	1.52	3.73	1.50	3.89	6.09	0.97	5.11	1.47	1.17	0.63
Dopamine D3	4.11	4.71	1.21	2.16	6.01	1.96	3.28	6.34	5.65	2.72
Dopamine D4	3.00	4.04	1.63	3.27	5.18	2.81	3.71	4.03	4.38	3.17
Epoxide hydrolase	1.27	4.30	1.06	1.72	3.78	2.83	2.65	4.38	3.35	4.31
Estrogen receptor beta	2.57	2.70	0.81	1.87	2.10	0.97	1.43	1.94	1.37	1.73
FABP4	3.31	4.41	1.03	2.57	4.26	1.16	1.36	3.16	2.30	1.88
HDAC6	1.99	1.82	0.50	2.38	2.45	0.76	1.65	1.64	0.83	1.20
HSP90 α	3.90	3.20	1.78	3.01	6.67	1.03	3.33	3.30	4.02	2.25
Histamine H2	0.57	1.35	1.16	2.50	4.36	0.84	1.52	4.19	2.36	2.34
Histamine H3	1.04	2.17	0.83	3.77	2.06	1.20	3.12	3.57	1.21	2.20
MMP9	1.41	1.38	0.87	2.57	1.81	0.61	1.47	1.66	1.04	1.86
Muscarinic M2	2.48	4.23	1.45	3.96	3.31	2.19	5.12	4.92	2.41	4.16
Muscarinic M3	3.80	3.61	1.96	3.02	4.98	4.97	6.62	6.71	5.03	4.06
PPAR γ	2.04	2.59	1.52	1.02	5.44	0.56	6.54	5.14	1.47	1.13
SIRT3	2.48	2.09	1.52	3.14	5.37	1.34	2.69	3.91	1.60	4.45
SYK	1.79	2.17	1.16	5.33	2.90	0.80	1.20	3.26	1.16	1.24
α_1 A adrenergic	4.62	3.68	1.12	1.79	3.26	2.70	3.82	5.36	2.78	1.84
α_1 B adrenergic	2.01	4.45	1.50	3.87	5.53	1.93	4.62	6.13	4.66	3.56
β_1 adrenergic	4.95	4.18	1.50	5.24	6.84	1.67	3.64	3.66	3.74	3.83

These findings suggest that ImagiChem's image-guided molecular design can yield not only chemically valid but also biologically plausible scaffolds, capable of interacting with key therapeutic targets. The observed polypharmacological tendencies, especially in compounds **7** and **8**, underscore the algorithm's potential for generating chemotypes with multi-target profiles, a valuable feature in drug discovery for complex diseases.

Further validation through molecular dynamics simulations and experimental assays will be necessary to confirm these interactions in vitro, yet the present results provide compelling preliminary evidence. Overall, the IVS data reinforce the hypothesis that aesthetic and structural cues extracted from artworks can drive the creation of novel, pharmacologically relevant molecular entities, bridging creativity and medicinal chemistry.

Robustness under Gaussian image perturbation

To assess the contribution of visual structure to molecular generation, we investigated how progressive Gaussian degradation of an input image affects the molecular output of our Image→SMILES generator. We compared distributions of physicochemical properties and structural similarity metrics across five noise conditions (original, $\sigma = 5, 25, 75, 100$). Using *The Birth of Venus*, selected for exhibiting the highest proportion in the DBPP score, as a reference input, we generated 1,307 compounds. We observed only minor variations in the global physicochemical distributions (mean molecular weight: 476.21 $\rightarrow$ 482.85 on average, from the original, with $\sigma = 100$). Properties more sensitive to polarity exhibited systematic but moderate shifts: mean cLogP decreased from 2.605 to 1.527, while PSA increased from 127.06 to 155.56 (original $\rightarrow \sigma = 100$) (Figure S1). These modest changes are expected, as the codec intrinsically encodes the permissible range of physicochemical values within its latent representation, thereby constraining their variability even under image perturbation. In contrast, structural comparisons revealed more potent effects. Pairwise fingerprint similarity between molecules generated from original versus blurred inputs yielded a median Tanimoto coefficient of ~ 0.46 across blur levels ($\sigma = 5\text{--}100$) (Figure S2 and Table S2), while the fraction of identical SMILES was effectively zero. Likewise, scaffold overlap (Bemis–Murcko) was negligible (median scaffold Jaccard = 0.0),

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3 indicating that Gaussian degradation typically produces alternative scaffolds rather than perturbed
4 analogues of the same core (Table S2). Despite these structural divergences, SA scores and PAINS
5 incidence remained effectively stable (mean $\Delta SA \approx$ of a few 10^{-2} units; PAINS $\sim 1\%$ across
6 conditions) (Table S2). Taken together, these findings indicate that while aggregate
7 physicochemical property distributions are partially conserved, reflecting the codec's intrinsic
8 representation of valid chemical space, the generative mapping from image to canonical scaffold
9 is sensitive to moderate image degradation and tends to yield structurally distinct molecules rather
10 than close analogues. All analyses were implemented in Python using RDKit for cheminformatics
11 processing; full per-compound results and analysis scripts are provided in the Supporting
12 Information to ensure reproducibility.
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28 DISCUSSION

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31 ImagiChem has launched a new paradigm in computational chemistry, addressing a fundamental
32 limitation of DNDD methods: the inherent bias of in-depth generative models toward known data
33 distributions, which limits their serendipity and the exploration of remote chemical spaces. We
34 have reintroduced algorithmic serendipity by using visual art, an input rich in structure but
35 detached from chemical precedents, as a structured prior for molecular generation.
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42 ImagiChem's distinctive strength lies in its entirely deterministic nature. Unlike stochastic
43 creative AI systems, the algorithm guarantees full reproducibility of each molecule, traceable to a
44 single integer seed cryptographically derived from the input image. This principle is crucial to
45 aligning with the required standards of scientific rigor.
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51 Applying the pipeline to a corpus of ten artworks has demonstrated its extraordinary
52 effectiveness as an engine of aesthetic chemical creativity. Regardless of size or pictorial style,
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ImagiChem generated molecules with an exceptional novelty rate, averaging 99.17% against the PubChem database, confirming its ability to explore unexplored chemical regions. In particular, the algorithm demonstrated a rediscovery capability: a molecule generated from the work *Virgin Annunciate*, DEABL (an approved anticonvulsant), was identified, demonstrating that aesthetic priors can converge on pharmacologically significant scaffolds.

Pharmacological evaluation strengthened the biological potential of the generated compounds. Our study identified the DBPP predictor as the most reliable model for discriminating against drug-like compounds. Applying this threshold (0.736), it was found that, on average, 13.55% of the generated molecules fell within the drug-candidate range. Interestingly, the distribution of molecules with high DBPP scores was heterogeneous, with works such as *The Birth of Venus* producing the highest proportion (28.77%), suggesting that aesthetic complexity and textual density positively correlate with molecular intricacy and aromaticity.

Even works of contrasting texture and color composition, such as *Vucciria* and *The Scream*, maintained a high level of novelty, reflecting ImagiChem's robustness across visual domains.

IVS done on the top ten *Vucciria* compounds provides compelling preliminary evidence of biological relevance. Several candidates showed predicted affinities above the CNN_VS threshold of 6.3 for key therapeutic targets. Compound **4** showed high affinity for the adenosine A_{2A} receptor, while compounds **7** and **8** demonstrated multi-target potential, interacting with the muscarinic M3 receptor.

Despite these promising preliminary results, rigorous experimental validation, including molecular dynamics simulations and in vitro assays, is essential to confirm the predicted ligand-target interactions.

The Gaussian blur analysis shows that while the Image→SMILES model reliably maintains overall physicochemical properties, it struggles to reproduce molecular scaffolds when the input image is degraded consistently. A median fingerprint similarity of about 0.46 indicates that Gaussian perturbations typically alter core structures rather than minor substituents. This suggests two interpretations: (1) for tasks requiring faithful scaffold preservation, additional architectural or training constraints (e.g., scaffold-conditioned decoding or contrastive consistency) are needed; (2) for diversity-oriented design, the model's sensitivity can be leveraged, image blur effectively promotes scaffold exploration without compromising drug-likeness or synthetic accessibility. The choice between robustness and diversity should thus be made explicitly and reflected in both training and evaluation protocols.

Looking to the future, ImagiChem provides a conceptual foundation for the evolution towards multimodal generative architectures. Future improvements will focus on integrating joint image-molecule embeddings (e.g., via CLIP-type encoders and Transformer-based decoders), implementing explainable attention mechanisms that directly link visual features to specific molecular substructures, and using reinforcement learning to optimize scaffolds for drug targets while preserving their aesthetic origins.

In conclusion, ImagiChem demonstrates that non-scientific human artifacts, works of art, can serve as a creative and structured complement to rational drug design. The combination of high molecular novelty, the rediscovery of existing drugs, and the identification of pharmacologically plausible candidates marks the inauguration of a new field: aesthetic chemistry, where the compositional harmony of art algorithmically merges with molecular functionality.

METHODS

1. Overview of the Computational Workflow

The software takes a grayscale image as input and, for each row of pixels, produces an “extended” string of atomic symbols that is subsequently converted into a molecular graph. The molecule is then assembled from a curated library of cores and functional groups, converted to an RDKit Mol object, sanitized, and scored.

The main computational steps are:

1. *Seed generation.* A deterministic random seed is derived from the SHA-256 hash of the grayscale image bytes (`image_to_seed`). This ensures complete reproducibility of all random operations (`random.seed`, `numpy.random.seed`, `RNG.seed`).
2. *Image processing.* The image is read in grayscale and split into rows of pixels (`split_pixel_rows`).
3. *Row-level molecular encoding.* For each row:
 - Row statistics are computed (`PixelLineAnalyzer.analyze_pixel_pattern`).
 - These statistics are mapped to atomic counts (`PixelLineAnalyzer.generate_molecular_formula_string`).
 - A molecular assembly is generated from the resulting string (`assemble_from_input_string`).
 - The structure is converted to an RDKit molecule, sanitized, and 2D coordinates are computed.
4. *Post-processing.* The generated SMILES strings are refined (`refine_smiles`), and the SA score is computed (`calculate_sa_score`, rescaled to 0–10). Pan-assay Interference Compounds (PAINS) filters are also applied for a druglikeness evaluation.

5. *Output.* The molecules are ranked by increasing SA, and a list of (SMILES, SA score, PAINS filters) tuples is returned.

2. Mapping from Pixel Statistics to Atomic Counts

All numerical mappings are implemented in `PixelLineAnalyzer.generate_molecular_formula_string` and `_map_value`.

The exact mathematical forms, constants, and rounding rules are summarized below.

2.1. Linear Mapping Function

Each numeric feature v is linearly mapped from $[in_min, in_max]$ to $[out_min, out_max]$ according to:

$$\text{map}(v) = (v - in_{\min}) \cdot \frac{out_{\max} - out_{\min}}{in_{\max} - in_{\min}} + out_{\min}$$

If $in_max == in_min$, the function returns out_min to prevent division by zero.

2.2. Constants and Intervals

- Carbon count (C):
 - `c_range = (15, 40)`
 - `c_count = int(_map_value(mean, 0, 255, 15, 40))`
 - Clipped to `[15, 40]` using `np.clip`.
- Approximate total atom count:
 - `total_atoms_approx = int(_map_value(contrast, 0, 255, 10, 80))`
- If `other_atoms_max = total_atoms_approx - c_count < 5`, the carbon count is recomputed as `c_count = max(15, int(total_atoms_approx * 0.5))`
- Element-specific ranges:

Element	Range (min, max)
O	(1, 10)
N	(1, 8)
S	(0, 2)
F, Cl, Br, I	(0, 1)

2.3. Element-Specific Rules

- Oxygen (O):

count = int(_map_value(mean, 0, 255, 1, 10)), clipped to [1, 10].

- Nitrogen (N):

count = int(_map_value(std, 0, 100, 1, 8)), clipped to [1, 8].

(Note: the row standard deviation is mapped from [0, 100] rather than [0, 255].)

- Sulfur (S):

Binary rule: count = 1 if features['num_peaks'] > 250, otherwise 0.

- Halogens (F, Cl, Br, I):

Set to zero by default. Every 20 rows ($i \% 20 == 0$), halogens may be added using count = rng.randint(min_val, max_val) and clipped to the specified range.

If adding an element exceeds total_atoms_approx, the count is reduced so that the total does not exceed this limit (while respecting each element's minimum count).

2.4. Final Atomic Sequence

The atomic symbols are concatenated according to their counts ([atom] * count), shuffled via rng.shuffle(formula_parts), and joined into a single continuous string (e.g., "CCCC...").

2.5. Rounding and Fraction Handling

All floating-point results are truncated to integers using `int()`, corresponding to truncation toward zero. No stochastic rounding is applied. The mapping process is entirely deterministic given a fixed seed.

3. Row Aggregation Strategy

Rows are processed independently.

Each pixel row yields its own atomic pool and molecular assembly. There is no global atom pool accumulated across rows. This design is implemented in `run_imagichem_processing`, which iterates over rows, constructs each molecular formula, and processes them separately.

4. Curated Libraries of Cores and Synthons

The provided `groups.py` file defines a minimal example library (`GROUP_LIBRARY`) containing six functional groups (ester, amide, amine, alcohol, nitro, ether). Each entry includes:

- a factory function generating a `MoleculeGraph` fragment,
- atomic requirements,
- and a probabilistic weight.

The `cores.py` file provides the foundational molecular cores of a curated fragment library, including over 1,500 single-pharmacophore fragments from Enamine.

The use of fragments bearing a single pharmacophore (rather than multiple, distally separated functional groups) is advantageous: this simplifies the binding interaction to one key moiety, allowing (i) higher synthetic tractability of fragment growth (fewer conflicting vectors and functional groups to optimise) and (ii) more straightforward interpretation of structure-activity relationships and binding mode evolution.

From a fragment-based drug discovery (FBDD) perspective, designing a library around such simplified, single-pharmacophore entities is aligned with best practice: fragments are by definition low in molecular weight, low in complexity (following e.g. the “Rule of Three”: MW <300, ≤ 3 H-bond donors, ≤ 3 H-bond acceptors, clogP ≤ 3) and yet high in atom-efficiency, thereby enabling a small library to sample chemical space efficiently.

The Enamine library further ensures that all compounds pass strict “Ro2-like” physicochemical and structural filters (excluding PAINS, reactive, and toxic motifs), making initial hits less likely to be false positives and synthetically tractable for follow-up.

By anchoring on such well-defined, stable fragments and scaffolds, the workflow facilitates a “hit to lead” transition in which the fragment core is elaborated (via fragment growing, merging, linker insertion, or scaffold extension) in a controlled manner, preserving the original binding moiety, while adding vectors for potency, selectivity, and ADMET optimisation. The literature shows that high-quality fragment libraries (with careful attention to diversity, pharmacophore novelty, and synthetic follow-up potential) are critical to the success of FBDD campaigns, especially for challenging targets²³.

In short, the fragment library gives a chemically validated seed (a fragment with a known drug-like core and a single binding pharmacophore), the scaffold provides the structural backbone, and from there, a medicinal chemistry trajectory can be laid out in a modular, rational fashion.

5. Parameters of the Hierarchical Assembly

5.1. Core Selection and Insertion

- Core selection:

Cores whose requirements are satisfied by the atom pool are identified, and one is selected via `RNG.choice(candidates)`.

- Number of cores (`n_cores`):

Defaults to 1; if `sum(pool.values()) >= 12`, `n_cores = RNG.choice([1, 2])`.

- If a second core is added, the algorithm searches for a host atom with at least one available valence and not part of the first core. If unavailable, a carbon linker is introduced (`needed_atoms['C'] += 1`).

5.2. Main Assembly Loop

The main loop runs while `sum(pool.values()) > 0`.

Each iteration:

- Identifies eligible functional groups whose atomic requirements are met.
- Each group has an independent insertion probability given by its weight (e.g., 0.45 for esters and amides).
- If a group is selected, an attachment site (host) is chosen using `choose_host(g)`, and the group is merged into the graph. Atomic requirements are subtracted from the pool.

If no group is selected, a fallback rule adds a single atom chosen by `choose_next_elem_for_chain(pool)`, which determines the element with the highest valence still available.

5.3. Host Selection

Hosts are chosen from atoms with remaining valence (`valence_rem ≥ 1`), sorted by (number of neighbors, `-remaining_valence`) to prefer less substituted sites.

A random selection is made among the top 50% (up to 6 candidates).

5.4. Double-Bond Demotion and Valence Relaxation

The function `demote_some_double_bonds_until_capacity(g, needed_hosts)` converts double bonds (preferentially involving carbon) to single bonds until enough attachment sites are available.

This function is invoked when the number of available hosts is insufficient for adding new fragments.

5.5 Final Connectivity

Disconnected components (`components(g)`) are connected iteratively.

Up to 200 attempts are made to join small components through available host atoms. Each successful connection consumes one unit of valence from each atom involved.

6. RDKit Conversion and Sanitization

6.1. Conversion

`graph_to_rwmol(g)` builds a mutable RDKit molecule (`Chem.RWMol`) by adding atoms and bonds (`BondType.SINGLE`, `DOUBLE`, `TRIPLE`).

6.2. Sanitization

- Default RDKit sanitization:

`Chem.SanitizeMol(mol)` is called with default flags.

- Fallback:

If sanitization fails, all double bonds are demoted to single bonds (`g.demote_bond`), and the molecule is rebuilt and sanitized again.

In the present study, stereochemical configurations were not assigned to the generated compounds. This choice was deliberate and reflects both methodological and conceptual considerations. The primary objective of the generation process was to explore topological and

pharmacophoric diversity rather than to model specific three-dimensional conformations. Since stereochemistry does not alter the two-dimensional connectivity of atoms but merely defines their spatial arrangement, omitting stereochemical information allows the system to operate within a simplified, yet chemically meaningful, representation of molecular space.

Furthermore, the construction procedure relies on fragment-based assembly and topological rules, without explicit evaluation of three-dimensional energies or conformational optimization. Under such conditions, any stereochemical assignment, whether random or deterministic, would be arbitrary and unsupported by the underlying physical model. Including stereocenters without energetic or conformational validation could lead to ambiguous or chemically inconsistent results.

Finally, by excluding stereochemical specification, the generated SMILES remain canonical and unambiguous, facilitating downstream processing, comparison, and validation of molecular structures. Stereochemistry can be introduced at a later stage, if required, through dedicated conformational analysis or stereoselective enumeration once energetically and pharmacologically relevant configurations can be meaningfully defined.

2D coordinates are generated using `AllChem.Compute2DCoords(mol)` for visualization purposes only.

7. Probabilistic and Numerical Parameters

Parameter	Description	Value/Range
Image seed	SHA-256 hash of image bytes	Deterministic
Mapping mean→C	[0,255] → [15,40]	Linear
Mapping contrast→total atoms	[0,255] → [10,80]	Linear
Mapping mean→O	[0,255] → [1,10]	Linear

Mapping std→N	[0,100] → [1,8]	Linear
Peak threshold	_find_peaks/_find_valleys	0.3
S presence rule	num_peaks >250	Binary
Halogens	allowed every 20th row	0–1
Group insertion probability	weight GROUP_LIBRARY	in 0.15–0.55
Max connection attempts	200	
Number of cores	1 or 2	depends on the atom pool size

8. RDKit and Library Versions

The program requires RDKit (Chem, AllChem, SA_Score, Descriptors, Draw) and auxiliary packages (NumPy, OpenCV, PubChemPy).

For reproducibility, the specific version number of each is stated (RDKit 2025.9.1, NumPy 2.3.3, OpenCV 4.12.0.88, Pillow 10.2.0, PyQt6 6.9.1).

9. RDKit Sanitization

Molecules were sanitized via Chem.SanitizeMol in RDKit. The SanitizeMol function in RDKit attempts to validate and standardize chemical structures. This includes critical tasks such as: (i) determining aromaticity, (ii) calculating hybridization, and (iii) validating valence. If the SanitizeMol function fails, a recovery fallback is initiated to resolve valence issues and allow the crucial RDKit molecule sanitization step to be retried.

10. Molecular docking

Protein structures were prepared using YASARA (v. 25.1.13)²⁴ under physiological pH (7.4). Ligands' 3D structures were generated and minimized in Open Babel (v. 3.1.1),²⁵ reflecting

physiological pH (7.4) states. Docking simulations used GNINA (v. 1.3)²⁶ in rescoring mode, with CNN_VS scoring functions. Binding pockets were defined from co-crystallized ligands using AutoDock Tools (v. 1.5.7).²⁷

11. PubChem similarity search script

Searching PubChem is handled by the `pubchem_check.py` script, which implements a cheminformatics pipeline for batch structural screening of compounds against the PubChem (NLM) database. The system is configured to perform only exact structural matching. The core logic is encapsulated in the `get_pubchem_results()` function, which uses the `pubchempy` library to search for a direct match to the provided SMILES string. For each SMILES, the script returns the associated CID (Compound ID) if found in PubChem, or the NEW flag if the compound is not present in PubChem. Results are processed in batches from an input `.csv` file and written to a separate tab-delimited `.txt` file.

12. QED and DBPP calculations

QED and DBPP scores were computed using internal scripts provided inside the GitHub shared repository and the DBPP-Predictor GUI, respectively, using default settings. Similarity checks against PubChem employed `PubChemPy`.

13. Gaussian blur and statistical comparisons

To quantify structural and physicochemical sensitivity to image degradation, we compared distributions of MW, cLogP, cLogS, and PSA, and performed paired structural-similarity analyses across 1,307 compounds. SMILES validity, SA_score, and PAINS annotations were carried over

from the generator output. For structural similarity, we computed per-compound fingerprint-based Tanimoto similarities using RDKit fingerprints (RDKFingerprint for throughput; results are qualitatively consistent with Morgan/ECFP approaches) and extracted Bemis–Murcko scaffolds to measure scaffold overlap (Jaccard index). Paired statistical comparisons between original and blurred conditions used Wilcoxon signed-rank tests and paired t-tests where appropriate; effect sizes were assessed qualitatively in light of the large sample size ($n = 1,307$). We used thresholds commonly adopted in cheminformatics to interpret similarity (Tanimoto ≥ 0.85 for near-identical; 0.7–0.85 for close analogues; ≈ 0.4 –0.5 indicating moderate resemblance).

ASSOCIATED CONTENT

Data availability

All data supporting this study are publicly accessible on Zenodo (<https://zenodo.org/records/17549410>).

Code availability

The ImagiChem code and generated molecules are available on GitHub (<https://github.com/rocco-b/ImagiChem>). Furthermore, to ensure full reproducibility and platform independence, given that minor variations can arise from differences in library or interpreter versions, we have compiled the entire codebase into standalone executables for both Windows and Linux (ImagiChem-1.0-win64.msi and ImagiChem-1.0-linux, respectively), which are also available in the project repository.

Supporting Information. The Supporting Information is available free of charge at <https://pubs.acs.org/doi/>. Tables S1 and S2, Figures S1 and S2.

The following file is available free of charge.

Supporting Information (PDF)

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Notes

The authors declare no competing interests.

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ImagiChem: Deterministic Image-Conditioned Generation of Chemically Valid and Drug-Like Molecules

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Table S1. This Table Summarizes the Protein Models Utilized During the IVS (Protein) Phase. For Each Model, the Identification Code (PDB ID) is Provided, Along with the Source Database from which it was Obtained: the Protein Data Bank (PDB) or PDB-REDO (REDO). Additionally, the Spatial Coordinates Defining the Center of the Docking Simulation Box (Center x, Center y, Center z) are Listed, as are the Dimensions of the Simulation Box Along Each Axis (Npts x, Npts y, Npts z). A Uniform Spacing Value of 1 Å was Applied Across All Simulation Boxes to Maintain Consistency in Grid Resolution

Protein	PDB ID	Source	Center x	Center y	Center z	Npts x	Npts y	Npts z
Adenosine A _{2A}	5OLH	REDO	-21.392	11.064	15.644	22	32	26
Tyrosine-protein kinase ABL2	2XYN (C)	PDB	-63.101	10.422	-34.090	20	22	18
AKT2	3D0E (A)	PDB	21.454	-20.504	8.109	30	26	26
Acetyl-CoA carboxylase 2	3JRX	PDB	-45.861	24.596	-18.365	22	24	24
Acetylcholinesterase	6O4W (B)	REDO	89.919	94.443	17.303	28	24	20
BACE-1	4DJW (B)	REDO	16.915	30.739	57.257	30	24	22
Carbonic Anhydrase II	4HT2 (A)	REDO	-3.826	24.016	-4.778	26	20	24
Cyclin-dependent kinase 2	1KE9	PDB	-9.642	48.089	9.126	22	20	24
Dopamine D3	7BVQ (B)	PDB	3.533	18.093	-15.119	20	26	24
Dopamine D4	5WIV	PDB	-17.723	15.366	-18.918	22	24	24
Epoxide hydrolase	3WKE	PDB	-16.428	-4.880	68.334	22	22	26
Estrogen receptor beta	4J24	PDB	-25.059	39.562	-15.778	40	30	40
FABP4	7FXP	PDB	5.808	-5.026	-18.748	20	26	20
HDAC6	5EDU (B)	REDO	0.053	9.322	5.058	24	22	20
HSP90 α	4O09	REDO	2.893	-7.723	-24.333	22	20	24
Histamine H2	8YN3	PDB	96.425	113.687	73.330	20	20	24
Histamine H3	8YN5	PDB	83.635	127.608	125.033	24	22	22
MMP9	6ESM	PDB	1.365	50.745	19.807	24	20	22
Muscarinic M2	5YC8	PDB	182.291	26.539	523.962	24	20	26
Muscarinic M3	8E9W	PDB	132.204	126.493	157.463	20	22	24
Peroxisome proliferator-activated receptor gamma	4EMA	PDB	16.875	7.580	44.931	24	20	24
SIRT3	4JSR	REDO	26.037	46.249	-1.103	18	18	20
SYK	3EMG	REDO	17.324	40.930	9.178	22	30	26
α 1A adrenergic	8THL	PDB	149.183	139.195	166.735	24	20	22
α 1B adrenergic	7B6W	PDB	5.952	27.356	-11.503	24	20	24
β 1 adrenergic	7BVQ	PDB	18.718	-23.726	87.090	26	24	32

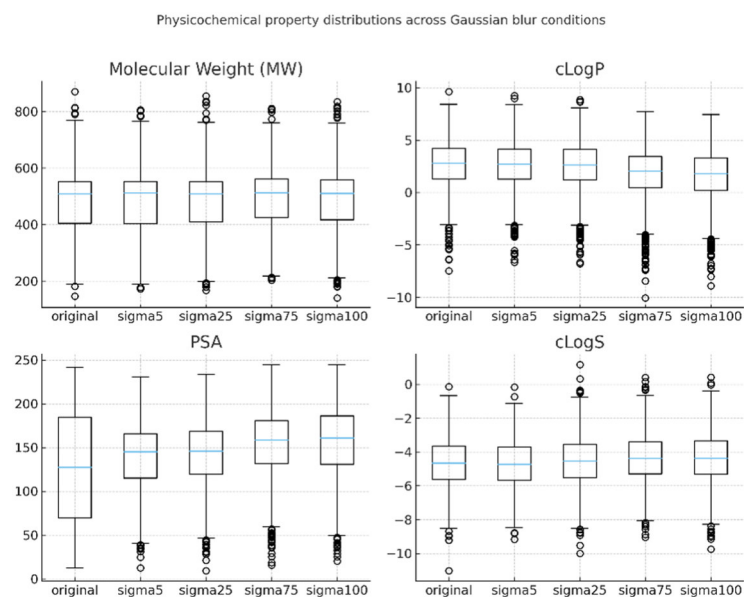


Figure S1. Physicochemical property distributions for molecules produced from original and Gaussian-degraded images. Boxplots show median and interquartile ranges for MW, cLogP, PSA, and cLogS across original and four Gaussian blur strengths (σ). Large blurs ($\sigma = 75$ and 100) systematically decrease cLogP and increase PSA relative to the original inputs.

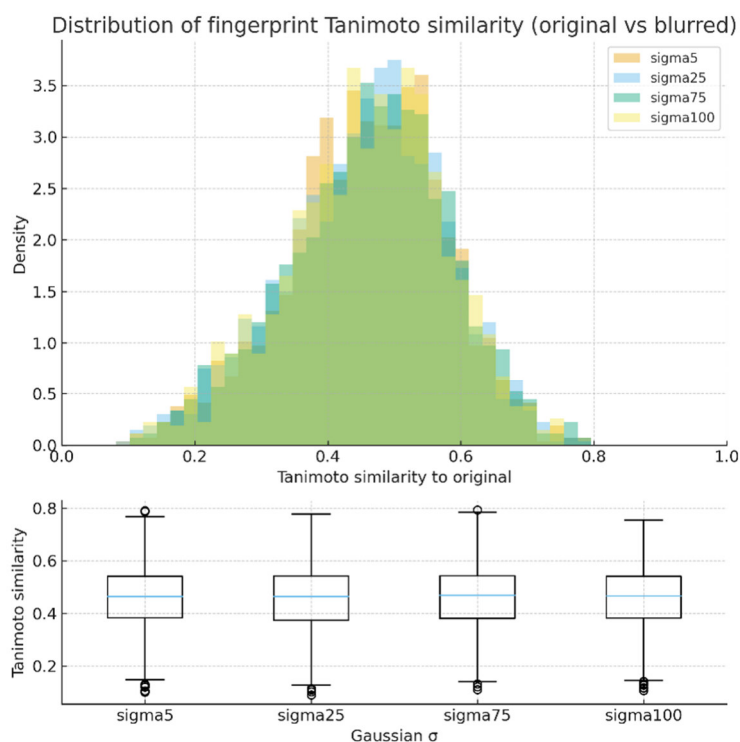


Figure S2. Structural similarity between original and blurred outputs. (Top) Kernel density estimates of per-compound fingerprint Tanimoto similarity (original vs blurred). (Bottom) Boxplots of the identical Tanimoto distributions. A median similarity of ~ 0.46 indicates moderate resemblance across conditions.

Table S2. Summary of Structural and Chemotype-Safety Metrics Comparing Original Outputs to Blurred Conditions

Condition	Median (orig vs cond)	TanimotoMean Tanimoto	Fraction identicalSMILES	Median Jaccard	scaffoldMean (cond – orig)	Δ SA scorePAINS fraction (cond)
$\sigma = 5$	0.4645	0.4578	0.00	0.00	-0.0125	1.45%
$\sigma = 25$	0.4640	0.4540	0.00	0.00	+0.0040	0.84%
$\sigma = 75$	0.4690	0.4606	0.00	0.00	+0.0597	0.99%
$\sigma = 100$	0.4668	0.4577	0.00	0.00	+0.0410	0.99%

(PAINS fraction: percentage of molecules flagged as PAINS under the condition; mean Δ SA score computed across compounds with available SA score.)